



An Open-Source Hybrid System Integrating PubMed Filtering, RAG, and LLM Models for Visualized Research Guidance

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Introduction

Help early-career researchers move from a broad curiosity to an evidence-grounded research project using AI technology.

- Undergraduates and novice investigators often have trouble developing research projects, despite having areas of interest
- Researchers are overwhelmed by the volume of information returned by online search engines
- The relationships between articles and their interests are not always clear
- Researchers often need to discuss (e.g., brainstorm) ideas before they can develop a project with confidence

CuraLit: a local, open-source workflow that turns a keyword-filtered PubMed collection into an explorable research knowledge base.

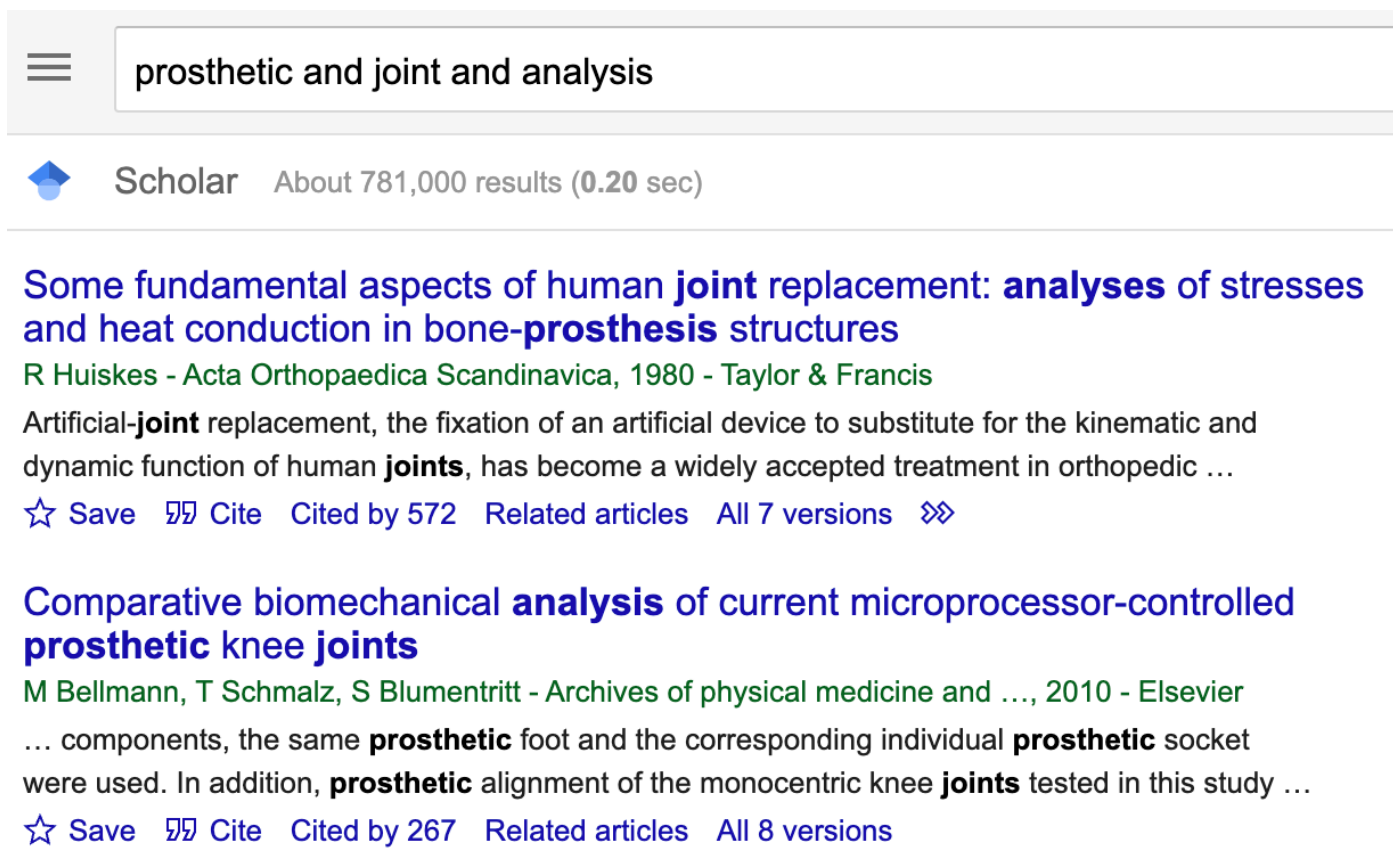
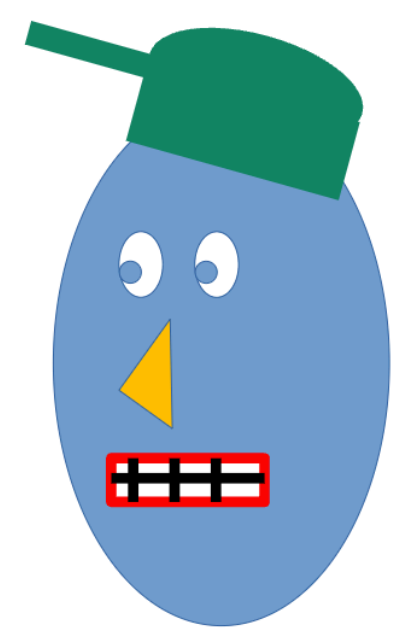
- Keywords related to an area of interest are parsed against the PubMed (<https://pubmed.ncbi.nlm.nih.gov/>) corpus to identify a subset of articles
- Analyze the effectiveness of keywords in retrieving articles from the PubMed corpus
- Visualize research trends with interactive plots
- Generate custom LLMs with Ollama or LM Studio for:
 - Answering questions about specific research domains
 - Brainstorming ideas and generating new insights
 - Choosing research topics and chatting about its design
 - Developing hypotheses
 - Synthesizing literature reviews

Features

A resource to help early-career researchers move from a broad curiosity to an evidence-grounded project

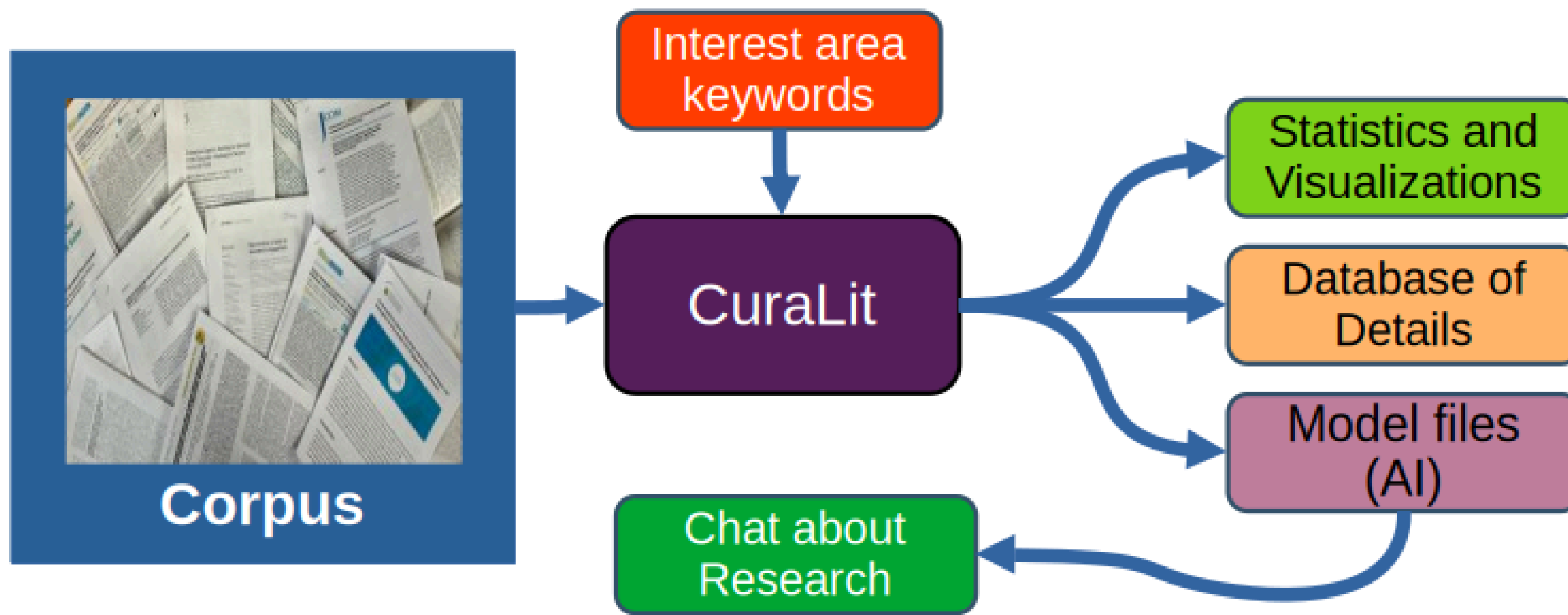
- Memory-Efficient Processing
 - Written in Rust to process millions of articles without performance degradation
- Flexible Keyword Matching
 - Search across PubMed (<https://pubmed.ncbi.nlm.nih.gov/>): titles, abstracts, MeSH terms, chemicals, and authors
- Checkpoint System
 - Continue searches from where you left off
- Statistical Analysis
 - Recommendations for keyword refinement
- Interactive Visualizations
 - Interactive HTML plots: heatmaps, scatter plots, and histograms to help researchers focus on specific searches
- LLM Generation
 - Generate Ollama model files for research conversations (<https://www.ollama.com>)
- RAG (Retrieval-Augmented Generation)
 - File-based Qdrant storage using Docker (<http://www.docker.com>); no separate server is required
- SQLite Database for Fact Verification
 - Database containing article details and citations to support fact-checking and reduce hallucinations

Traditional Searches Can Frustrate and Reduce Productivity



Indicated on the left, many students believe that success in research depends only on their skills to do the work of the project. They may not realize that the successes of their projects are initially based on the foundational work they do during development. CuraLit reduces the frustration of interpreting search-engine results by applying visual and AI-assisted guidance to help students navigate the literature and its concepts. Shown right is a typical keyword search with nearly 800,000 results which may lead students to feel overwhelmed.

Flowchart

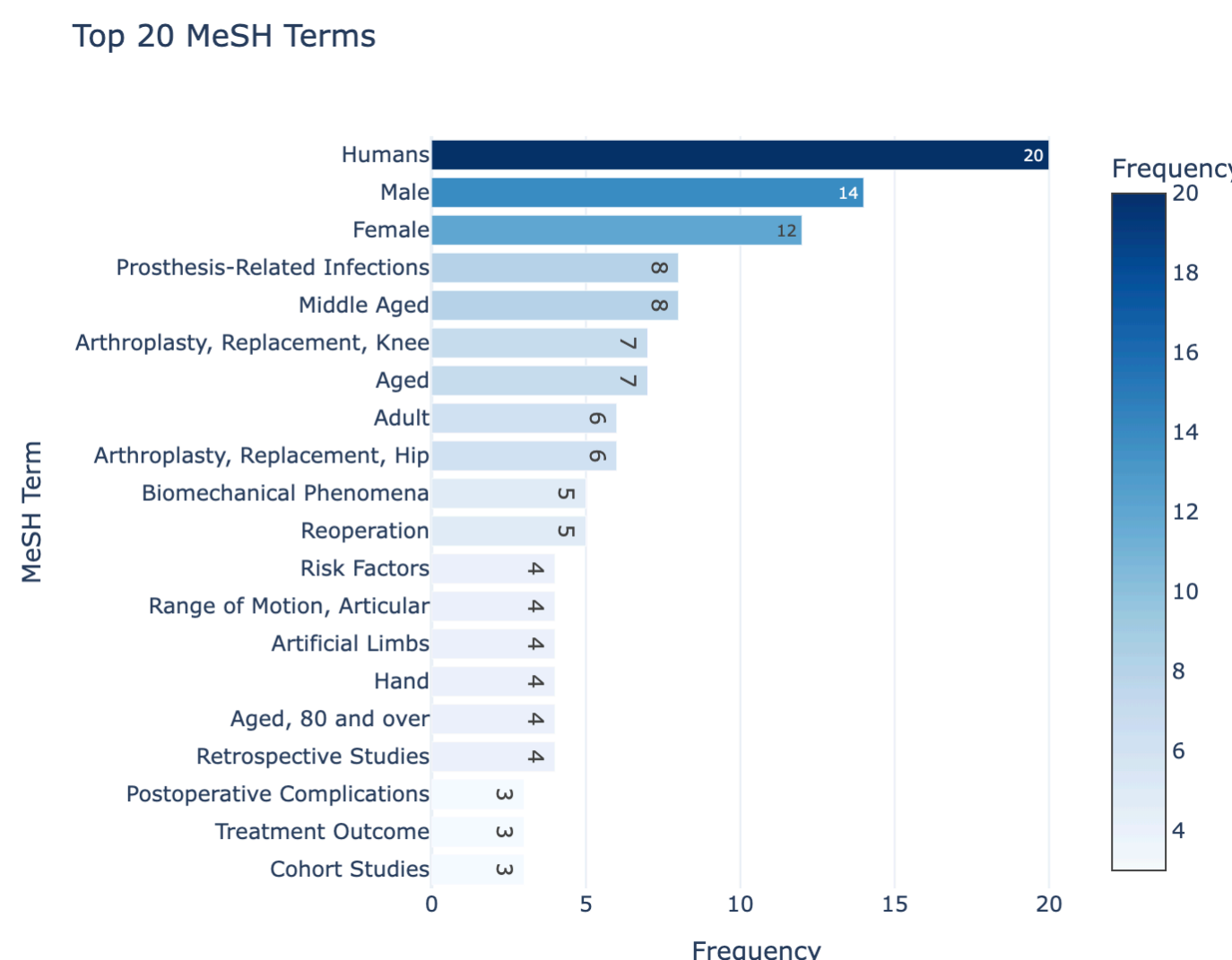
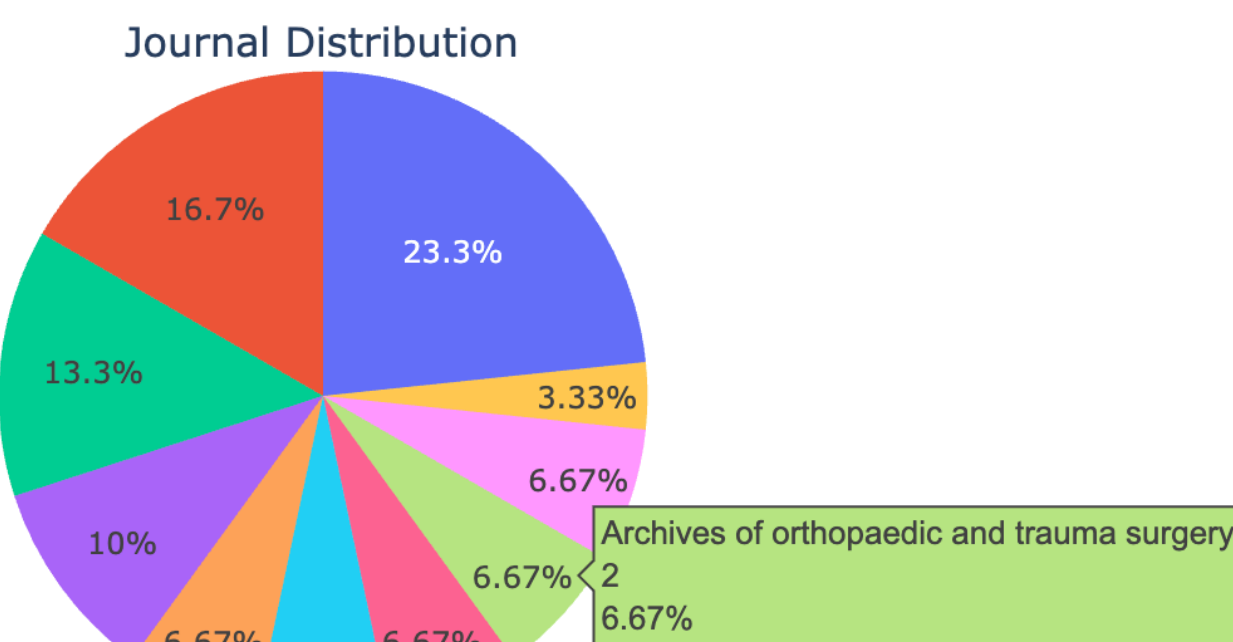


Project workflow: CuraLit filters a PubMed corpus according to user-selected keywords. It then builds a database from article details, methods, citations, and authors, and uses that knowledge to create model files for AI tools. Investigators can chat with the model through Ollama to explore concepts and develop an original research project. Statistics and visualizations provide feedback on keyword effectiveness, while the local database supports fact-checking.

CuraLit Steps

- 1. Curate:**
 - Choose terms that express a research interest; CuraLit filters PubMed XML files into a focused article collection.
- 2. Characterize:**
 - Calculate article, journal, and MeSH-term summaries; generate editable visualizations.
- 3. Ground:**
 - Build a searchable local knowledge base from the selected articles.
- 4. Question:**
 - Retrieve relevant passages before an LLM drafts a response; display supporting PMIDs.
- 5. Verify:**
 - Check cited PMIDs and article metadata against a local SQLite database.

Keyword Effectiveness in Retrieval

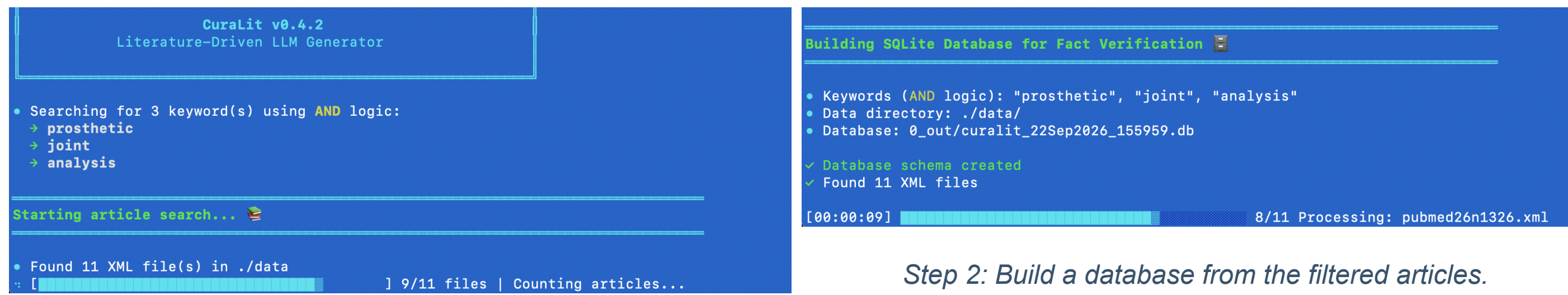


These visualizations guide keyword selection. The pie chart (left) shows the leading journals among articles containing the selected keywords. The histogram (right) shows the most common MeSH terms. MeSH terms are standardized labels for biomedical concepts that support consistent searching. A separate author-frequency plot (not shown) also provides additional guidance.

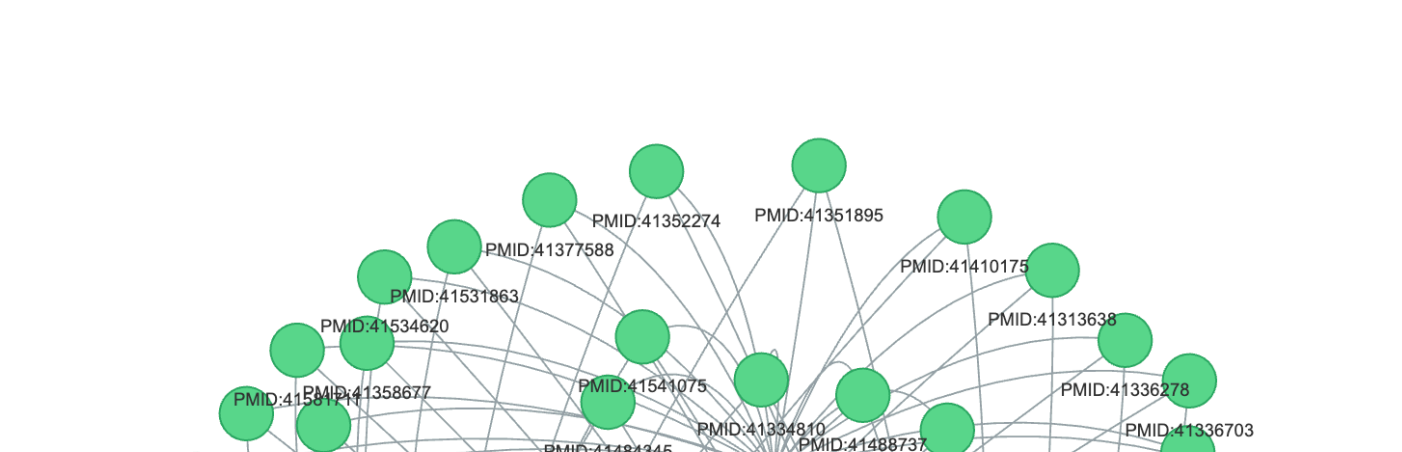
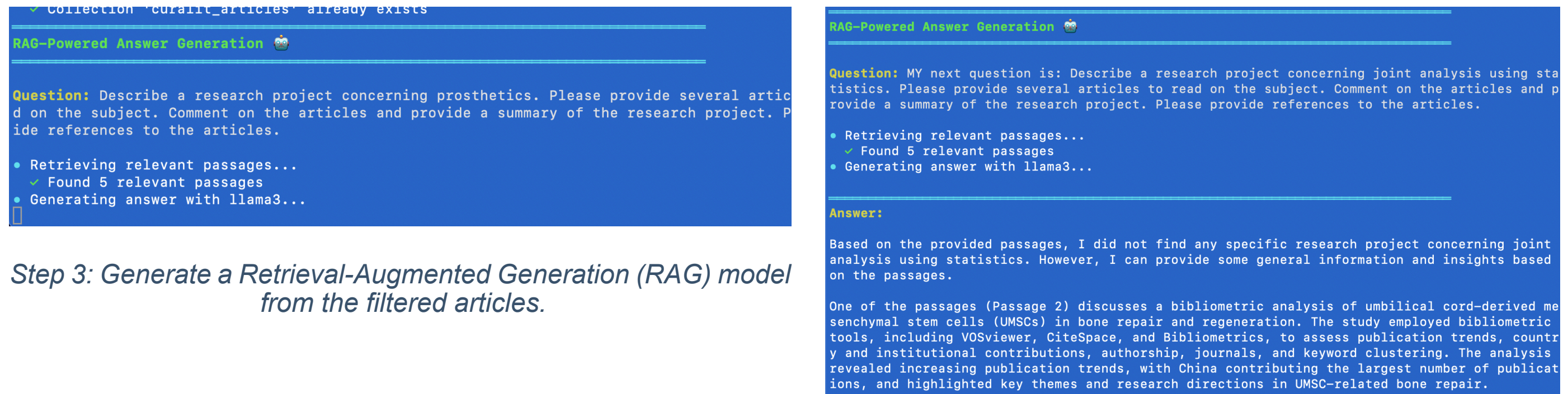
Visual Guidance for Project Development

- MeSH frequency:** Identifies the clinical, population, and methodological concepts most represented in the corpus.
- Journal distribution:** Provides a quick view of publication venues relevant to the topic.
- Knowledge network:** Links terms and articles to expose clusters, connections, and potential avenues for reading.
- Editable outputs:** Visualization scripts and data summaries can be adapted for a project, poster, or literature review.

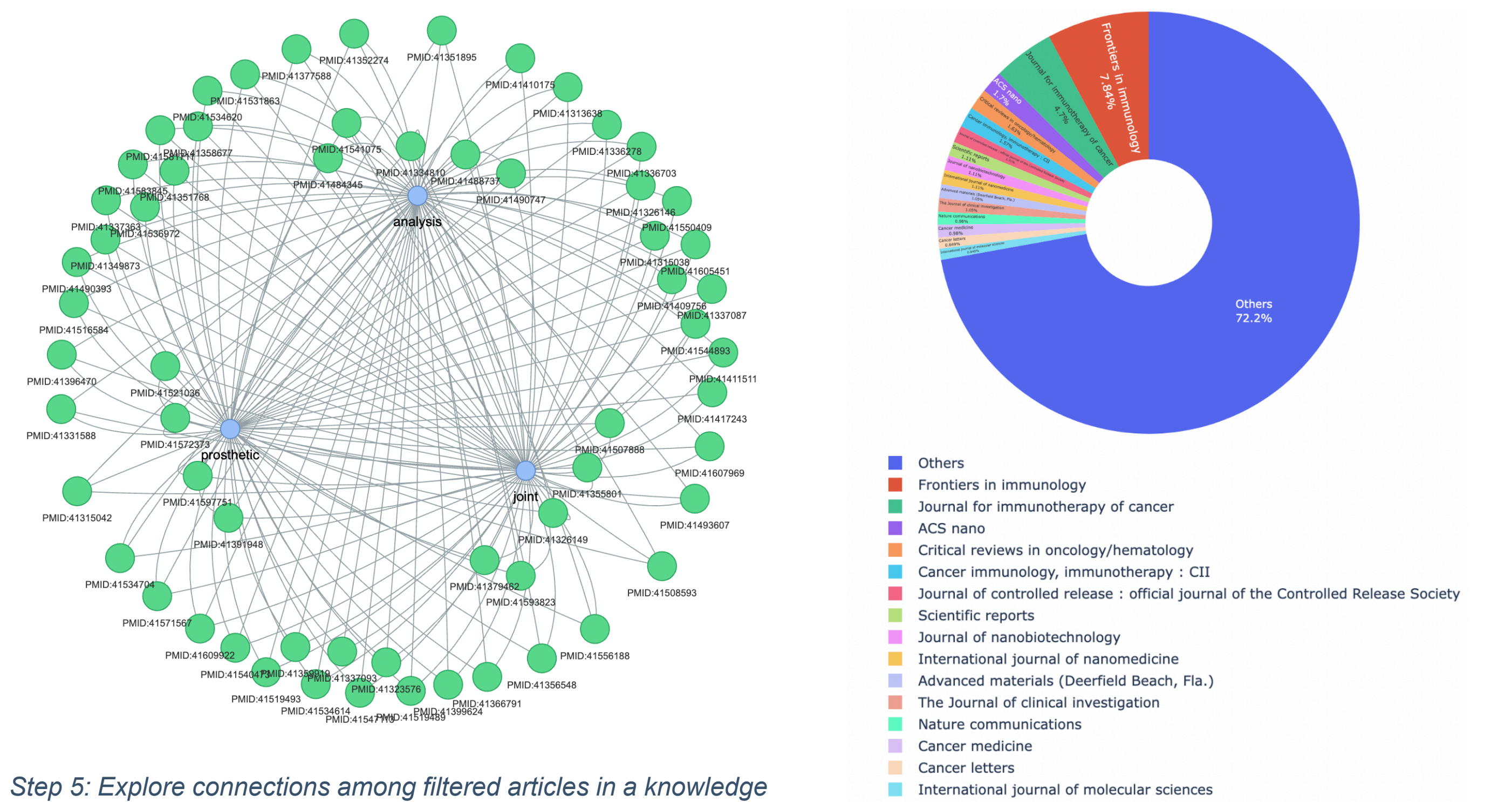
Launching the research process



Step 1: Select the PubMed XML files to filter by keyword.



Step 4: Ask a question and receive an answer from the RAG model.



Step 5: Explore connections among filtered articles in a knowledge network.

Step 6: View the distribution of filtered articles across journals.

Future Work

- Testing:** Test the tool to evaluate its effectiveness and usability.
- User experience:** Conduct user testing to evaluate ease of use and overall satisfaction.
- Resource consumption:** Optimize the CuraLit code and algorithms to reduce computational requirements.
- Novice researcher feedback:** Gather feedback from novice researchers to improve the tool's functionality and user interface.